Daniel DeLayo

Sixth-Year Computer Science Ph.D. Candidate

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Education

2021 – Present **Ph.D. Candidate, Computer Science**, Stony Brook University.
Expected graduation: *May 2027*
Advisor: *Michael A. Bender*

2017 – 2021 **B.S. Computer Science**, Stony Brook University.
Honors: *Summa Cum Laude, Honors College, Honors Computer Science.*
GPA: 3.95/4.0

Employment History

2019 – Present **Research Assistant**, Stony Brook University, *Computer Science.*

2021 – 2024 **Intern**, Sandia National Labs, *Cyber Security Algorithms Research.*

2018 **Research Assistant**, Stony Brook University, *Physics.*

Research Summary

My research focuses on how memory movement and contention limit performance, and on algorithms that overcome those limits. I develop parallel algorithms and space-efficient data structures that combine provable guarantees with high-performance implementations.

In caching, I study miss-ratio curves, caches with limited associativity, and caches with changing capacity. These curves were long considered too expensive to compute in production. As part of my MongoDB PhD Fellowship, I am integrating my algorithm into MongoDB, where it computes them with negligible overhead.

Beyond caching, I work on problems whose performance is dominated by memory movement: determinacy race detection in parallel programs, linear sketching for massive, dense graphs, and compact filters.

Research Publications

Journal Articles

1. D. Tench, E. West, V. Zhang, M. A. Bender, A. Chowdhury, **D. DeLayo**, J. A. Dellas, M. Farach-Colton, T. Seip, and K. Zhang. "GraphZeppelin: How to Find Connected Components (Even When Graphs Are Dense, Dynamic, and Massive)". In: *ACM Transactions on Database Systems* 49.3 (Sept. 2024), pp. 1–31. DOI: 10.1145/3643846.

2. J. Vorobyeva*, **D. DeLayo***, M. A. Bender, M. Farach-Colton, P. Pandey, C. A. Phillips, S. Singh, E. D. Thomas, and T. M. Kroeger. "Using Advanced Data Structures to Enable Responsive Security Monitoring". In: *Cluster Computing* 25.4 (Aug. 2022), pp. 2893–2914. DOI: 10.1007/s10586-021-03463-5.

Conference Proceedings

3. M. A. Bender, A. Conway, **D. DeLayo**, M. Farach-Colton, J. Han, L. He, R. Johnson, S. Kannan, W. Kuszmaul, D. Porter, and E. West. "Don't Melt Your Cache: Low-Associativity with Heat-Sink". In: *Proceedings of the 37th ACM Symposium on Parallelism in Algorithms and Architectures*. **Distinguished Paper Award**. ACM, July 2025, pp. 555–565. DOI: 10.1145/3694906.3743303.

4. J. L. Jones, A. T. Wilson, R. Gooding, **D. DeLayo**, K. Dalbey, J. Whetzel, and N. Sharan. "Behavioral Segmentation and Clustering of Geospatial Trajectories". In: *2025 IEEE 28th International Conference on Intelligent Transportation Systems (ITSC)*. IEEE, Nov. 2025, pp. 4375–4381. DOI: 10.1109/ITSC60802.2025.11423696.

5. D. Tench, E. T. West, K. Zhang, M. A. Bender, **D. DeLayo**, M. Farach-Colton, G. Gill, T. Seip, and V. Zhang. "Exploring the Landscape of Distributed Graph Sketching". In: *2025 Proceedings of the Symposium on Algorithm Engineering and Experiments (ALENEX)*. Society for Industrial and Applied Mathematics, Jan. 2025, pp. 133–146. DOI: 10.1137/1.9781611978339.11.

6. M. A. Bender, **D. DeLayo**, B. C. Kuszmaul, W. Kuszmaul, and E. West. "Increment-and-Freeze: Every Cache, Everywhere, All of the Time". In: *Proceedings of the 35th ACM Symposium on Parallelism in Algorithms and Architectures*. ACM, June 2023, pp. 129–139. DOI: 10.1145/3558481.3591085.

7. **D. DeLayo**[*], K. Zhang[*], K. Agrawal, M. A. Bender, J. W. Berry, R. Das, B. Moseley, and C. A. Phillips. "Automatic HBM Management: Models and Algorithms". In: *Proceedings of the 34th ACM Symposium on Parallelism in Algorithms and Architectures*. ACM, July 2022, pp. 147–159. DOI: 10.1145/3490148.3538570.

Preprints

8. Q. De Man, Q. Jafri, **D. DeLayo**, E. T. West, M. A. Bender, and D. Tench. *Fast and Compact Sketch-Based Dynamic Connectivity*. 2025. DOI: 10.48550/ARXIV.2509.14433.

[*]Joint first authors.

Software

2023 – Present **Increment-and-Freeze**, exact parallel LRU miss-ratio curves, up to 10× faster than the prior best parallel algorithm; being integrated into WiredTiger and OpenCilk.
`github.com/etwest/Increment-and-Freeze`

2025 **CUPCaKE**, dense-graph dynamic connectivity in 10× less memory than lossless systems.
`github.com/etwest/DynamicQueriesCC`

2024 **Landscape**, distributed graph sketching, ingesting up to 332M updates per second.
`github.com/GraphStreamingProject/Landscape`

2022 **GraphZeppelin**, out-of-RAM connectivity over 100× faster than Aspen and Terrace.
`github.com/GraphStreamingProject/GraphZeppelin`

GutterTree (pipeline hypertree), coalesces over 500M random updates per second.
`github.com/GraphStreamingProject/GutterTree`

Awards and Honors

2026 **MongoDB PhD Fellowship**.

HALG Invited Talk, Don't Melt Your Cache: Low-Associativity with Heat-Sink.

2025 **SPAA Distinguished Paper**, Don't Melt Your Cache: Low-Associativity with Heat-Sink.

2024 – 2025 **NSF Student Travel Grant**, SPAA.

2023 **CLSAC Invited Student Poster**, Chesapeake Large-Scale Analytics Conference.

2022 **GAANN Fellowship**, Stony Brook University.

Talks and Presentations

2026 **HALG (Invited)**, Don't Melt Your Cache: Low-Associativity with Heat-Sink.

MAPSP, Don't Melt Your Cache: Low-Associativity with Heat-Sink.

MongoDB (PhD Fellowship), Increment-and-Freeze: LRU Miss-Ratio Curves for Every Cache, Everywhere, All of the Time.

2025 **SPAA**, Don't Melt Your Cache: Low-Associativity with Heat-Sink.

NYU Theory Seminar, Increment-and-Freeze: Every Cache, Everywhere, All of the Time.

PhD Dissertation Proposal, Practical Sequential and Parallel Paging via LRU Miss Ratio Curves.

2024 **PhD Prelim**, Increment-and-Freeze: Every Cache, Everywhere, All of the Time.

2023 **CLSAC**, Stateful Streaming with External Memory On Workstations.

2022 **SPAA**, Automatic HBM Management: Models and Algorithms.

MAPSP, Automatic HBM Management: Models and Algorithms.

Teaching Experience

2025 **Distributed Systems**, Teaching Assistant.

Honors Algorithms, Probability Lecture.

2023 **Graduate System Security**, Teaching Assistant.

2021 **Honors Algorithms**, Teaching Assistant.

2018 **Intro to Computer Science**, Teaching Assistant.

Professional Service

2026 **Shadow Program Committee**, SPAA.

Artifact Evaluation Committee, ALENEX.

2025 **Shadow Program Committee**, SPAA.

Artifact Evaluation Committee, ALENEX.

2024 **Artifact Evaluation Committee**, ALENEX.

2023 **Peer Review**, ESA.

2022 **Peer Review**, IPDPS.